Given a function $f(x)$ that has positive $y-$values on a closed interval $[a,b]$, we measure the area that the graph of the function forms with the horizontal axis and the lines $x=a, x=b$ by evaluating the limit:

$$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x$$

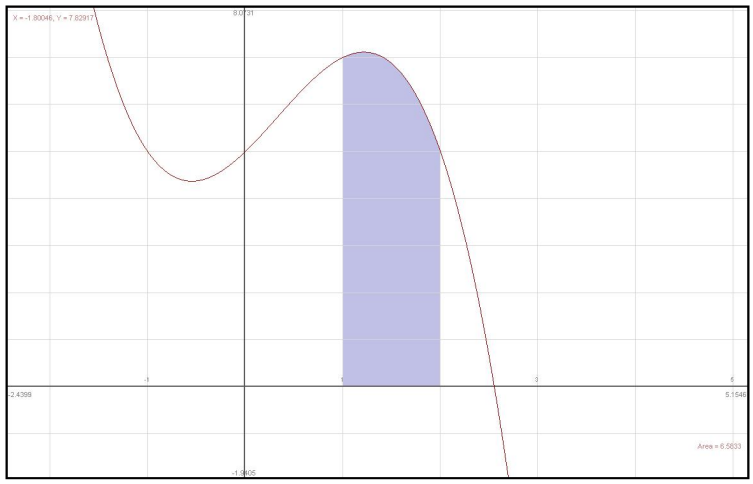


Figure: $f(x) = -x^3 + x^2 + 2x + 5 \quad [a,b] = [1, 2]$

Translating the Information	Properties of Sigma
$x_0 = a$ $x_n = b$ $\Delta x = \frac{b-a}{n}$ $x_1 = x_0 + \Delta x$ $x_2 = x_0 + 2\Delta x$ .... $x_i = x_0 + i\Delta x$ ... $x_n = x_0 + n\Delta x = a + n\frac{b-a}{n} = a + b - a = b$	$\sum_{i=1}^{n} k = (\underbrace{k + k + ... + k}_{n \text{ times}}) = nk$ $\sum_{i=1}^{n} (x_i \pm y_i) = \sum_{i=1}^{n} x_i \pm \sum_{i=1}^{n} y_i$ $\sum_{i=1}^{n} kx_i = k\sum_{i=1}^{n} x_i$ $\sum_{i=1}^{n} i = 1 + 2 + 3 + ... + n = \frac{n(n+1)}{2}$ $\sum_{i=1}^{n} i^2 = 1^2 + 2^2 + 3^2 + ... + n^2 = \frac{n(n+1)(2n+1)}{6}$ $\sum_{i=1}^{n} i^3 = 1^3 + 2^3 + 3^3 + ... + n^3 = \left(\frac{n(n+1)}{2}\right)^2$

Example:

$f(x) = (x-1)(2-x) = -x^2 + 3x - 2 \quad [a,b] = [1, 2]$

<u>Solution:</u>

$x_0 = a = 1$

$x_n = b = 2$

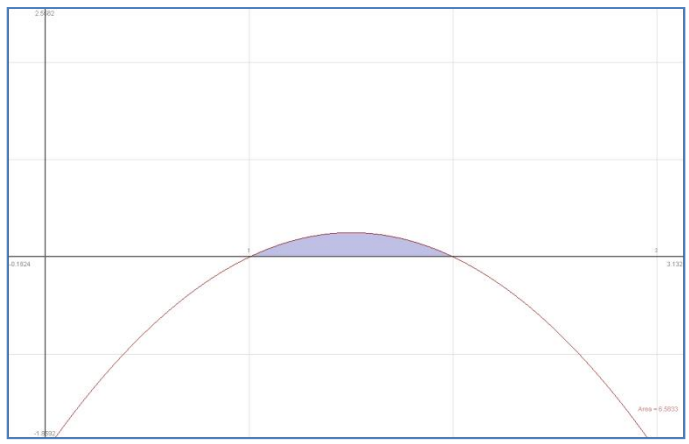
$\Delta x = \frac{b-a}{n} = \frac{2-1}{n} = \frac{1}{n}$

$x_i = x_0 + i\Delta x = 1 + \frac{i}{n}$

$f(x_i) = -\left(1+\frac{i}{n}\right)^2 + 3\left(1+\frac{i}{n}\right) - 2 = -1 - \frac{2i}{n} - \frac{i^2}{n^2} + 3 + \frac{3i}{n} - 2 = \frac{i}{n} - \frac{i^2}{n^2}$

Since $f(x) \geq 0$ when $x \in [1, 2]$, the area will be equal to the limit of the sums of rectangles as the number of rectangles (n) goes to ∞.

$$
\begin{aligned}
A &= \lim_{n\to\infty} R_n \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\Delta x \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} \left(\frac{i}{n} - \frac{i^2}{n^2}\right)\left(\frac{1}{n}\right) \\
&= \lim_{n\to\infty} \sum_{i=1}^{n} \left(i - \frac{i^2}{n}\right)\left(\frac{1}{n^2}\right) \\
&= \lim_{n\to\infty} \left(\frac{1}{n^2}\right)\sum_{i=1}^{n} \left(i - \frac{i^2}{n}\right) \\
&= \lim_{n\to\infty} \left(\frac{1}{n^2}\right)\left[\sum_{i=1}^{n} i - \sum_{i=1}^{n}\left(\frac{i^2}{n}\right)\right] \\
&= \lim_{n\to\infty} \left(\frac{1}{n^2}\right)\left[\sum_{i=1}^{n} i - \frac{1}{n}\sum_{i=1}^{n} i^2\right] \\
&= \lim_{n\to\infty} \left(\frac{1}{n^2}\right)\left[\frac{n(n+1)}{2} - \frac{1}{n}\left(\frac{n(n+1)(2n+1)}{6}\right)\right] \\
&= \lim_{n\to\infty} \left(\frac{1}{n^2}\right)\left[\frac{n(n+1)}{2} - \left(\frac{(n+1)(2n+1)}{6}\right)\right] \\
&= \lim_{n\to\infty} \left[\frac{n(n+1)}{2n^2} - \left(\frac{(n+1)(2n+1)}{6n^2}\right)\right] \\
&= \lim_{n\to\infty} \left[\frac{3n(n+1)}{6n^2} - \left(\frac{(n+1)(2n+1)}{6n^2}\right)\right] \\
&= \lim_{n\to\infty} \left[\frac{3n^2+3n}{6n^2} - \frac{2n^2+3n+1}{6n^2}\right] \\
&= \lim_{n\to\infty} \left[\frac{3n^2+3n-(2n^2+3n+1)}{6n^2}\right] \\
&= \lim_{n\to\infty} \left[\frac{n^2-1}{6n^2}\right] \\
&= \frac{1}{6}
\end{aligned}
$$



AREA = $\frac{1}{6}$ **square units**

Practice Problems

Using the limit definition of the area that the graph of a positive function forms with the horizontal axis and two vertical lines $x = a$ and $x = b$, find the exact area in each case:

1. $f(x) = (1-x)(x-2)(3-x) \quad [1,2]$
2. $f(x) = x - x^2 \quad [0,1]$
3. $f(x) = x^2 + x \quad [1, 4]$
4. $f(x) = -x^3 + 2x^2 \quad [0, 2]$
5. $y = 10 - x^2 \quad [1, 3]$